



Exploring Shared Bodily Control: Designing Augmented Human Systems for Intra- and Inter-Corporeality

Rakesh Patibanda
rakesh@exertiongameslab.org
Exertion Games Lab, Department of
Human-Centered Computing,
Monash University
Melbourne, Australia

Nathalie Overdevest
nathalie@exertiongameslab.org
Exertion Games Lab, Department of
Human-Centered Computing,
Monash University
Melbourne, Australia

Aryan Saini
aryan@exertiongameslab.org
Exertion Games Lab, Department of
Human-Centered Computing,
Monash University
Melbourne, Australia

Zhuying Li
zhuyingli@seu.edu.cn
Southeast University
School of Computer Science and
Engineering
Nanjing, China

Josh Andres
josh.andres@anu.edu.au
Australian National University
School of Cybernetics
Acton, Australia

Jarrold Knibbe
jarrod.knibbe@unimelb.edu.au
The University of Queensland
School of Electrical Engineering and
Computer Science Faculty of
Engineering
St Lucia, Queensland, Australia

Elise van den Hoven
Elise.VandenHoven@uts.edu.au
University of Technology Sydney
Sydney, Australia
Eindhoven University of Technology
Eindhoven, The Netherlands

Florian 'Floyd' Mueller
floyd@exertiongameslab.org
Exertion Games Lab, Department of
Human-Centered Computing,
Monash University
Melbourne, Australia

ABSTRACT

The human-computer interaction community has evolved from using body-sensing to body-actuating technologies, transforming the body's role from a mere input to an input-output medium. With body-sensing, the separation between the human and the computer is clear, allowing for an easy understanding of who is in control. However, with body-actuating technologies, this separation diminishes. These technologies integrate more closely with our bodies, where both the user and the technology can share control over their bodily interactions. In this workshop, we will explore this notion of sharing control, specifically focusing on experiences where users interact with their own bodies (intra-corporeal experiences), and interact with others using technology (inter-corporeal experiences). Our discussions and group activities will focus on brainstorming and designing within human augmentation, examining how this shared control can lead to innovative applications.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms.**

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

AHs 2024, April 04–06, 2024, Melbourne, VIC, Australia

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0980-7/24/04

<https://doi.org/10.1145/3652920.3653037>

KEYWORDS

Body-Actuating Technologies, Shared Control, Intra-Corporeal Experiences, Inter-Corporeal Experiences, Human Augmentation, Bodily Interactions, Wearable Robotics, Physical Computing, Sensory Augmentation, Input-Output Medium

ACM Reference Format:

Rakesh Patibanda, Nathalie Overdevest, Aryan Saini, Zhuying Li, Josh Andres, Jarrod Knibbe, Elise van den Hoven, and Florian 'Floyd' Mueller. 2024. Exploring Shared Bodily Control: Designing Augmented Human Systems for Intra- and Inter-Corporeality. In *The Augmented Humans International Conference (AHs 2024)*, April 04–06, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3652920.3653037>

1 INTRODUCTION

The domain of human augmentation is shifting from exclusively relying on body-sensing technologies towards embracing body-actuation technologies [33, 54], including but not limited to electrical muscle stimulation [2, 12, 23, 24, 39, 42, 45, 58, 59], exoskeletons [1, 20, 22, 26, 36, 52], and pneumatics [16, 48–50]. Body-sensing technologies conventionally position the human body as an input device, facilitating a clear understanding of control – where the distinction between human and computer-initiated actions is perceivable and evident [18, 29, 30]. In contrast, body-actuation technologies promote a sharing of bodily control between the user and a computational system. This integration redefines the body as a bidirectional medium of input and output while highlighting the body as the epicentre of the augmented experience [47]. This also blurs the once binary understanding of control [8, 9, 34], thereby challenging our perceptions of sense of agency – the feeling of

being in control of our actions [28]. We believe that investigating how body-actuating technologies alter our sense of agency is imperative for designing human augmentation systems that augment human capabilities and conserve user autonomy and control [20, 25, 33, 37, 53].

In this context, 'shared control' also resonates within the realm of play literature [38], particularly through the lens of 'Illinx' [10] – a state of relinquishing bodily control to foster deeper engagement with one's own body and those of others [11, 51]. Drawing from this perspective, we introduce 'intra-corporeality' and 'inter-corporeality' interactions as concepts for our workshop's proposed control-corporeality framework (Figure 1) [14, 17]. Intra-corporeality can be facilitated through the design of interactions enabled by technologies that integrate with an individual's body, promoting personal and introspective encounters with technology. In contrast, inter-corporeality can be facilitated through the design of interactions enabled by technologies that connect multiple individuals, thereby creating a shared system for bodily input and output. This approach not only fosters interpersonal interactions but also cultivates a collective experience and understanding. By centering our exploration on 'shared control' and 'corporeality,' we aim to delve into the design of augmented human systems through this control-corporeality framework.

In this workshop, we aim to use this framework, which is structured around two principal axes: the level of control and the spectrum of corporeality. The level of control (y-axis) ranges from a high level of human control to a high level of computer control, and the spectrum of corporeality (x-axis) transitions from intra-corporeal interactions to inter-corporeal interactions. This framework invites exploration of this continuum, which ranges from humans exerting a high level of control over their interactions to situations where computational systems take on a more significant level of control. By focusing on both intra- and inter-corporeal interactions, we aim to illuminate the multifaceted dynamics at play in designing augmented human systems. This workshop aims to explore this novel design space and provide practical learnings and insights to support future designs.

2 WORKSHOP GOALS

The workshop's objective is to transcend the conventional binary perspective on control within human-computer interaction, promoting a paradigm of shared control. The workshop endeavours to ignite innovative thinking and encourage cross-disciplinary partnerships among academics, designers, and technologists.

3 WORKSHOP STRUCTURE AND SCHEDULE

- **09:00 AM - 09:15 AM | Introduction & Welcome:** A brief session to welcome attendees, outline the workshop's objectives, and introduce the day's agenda.
- **09:15 AM - 09:30 AM | Opening Keynote:** A keynote presentation focusing on the critical concepts of shared control, intra-corporeality, and inter-corporeality, setting the theoretical groundwork for the workshop.
- **09:45 AM - 10:30 AM | Lightning Talk Presentations:** Participants will have 1 minute each to introduce themselves and discuss a system that exemplifies one of the quadrants

within the control-corporeality framework. This session aims to foster a diverse exchange of ideas and spark innovative thinking.

- **10:30 AM - 10:50 AM | Coffee Break:** A chance for participants to network, discuss ideas presented during the lightning talks, and prepare for group activities.
- **10:45 AM - 11:00 AM | Framework:** The control-corporeality framework is explained that participants will work with to design their systems.
- **11:00 AM - 12:00 PM | Group Activity:** In groups of four, participants will brainstorm and design augmented human systems within chosen themes such as sports, health, self-image enhancement, and games across the four quadrants of the control-corporeality framework. This activity will use brainwriting and design cards associated with the framework to encourage creative thinking and concept refinement.
- **12:00-12:30 | Presentation of Ideas:** Each group will be given five minutes to present their idea.
- **12:30-12:45 | Closing Remarks & Next Steps:** Concluding the workshop with a summary of key insights, discussions on potential collaborations, and outlining steps for future engagement and research dissemination.

To facilitate these activities, the workshop venue will be equipped with round tables for breakout discussions, a projector and AV system for presentations, power outlets for personal devices, and creative materials, including flip charts, post-it notes, and whiteboard markers.

4 POST-WORKSHOP PLANS

In collaboration with interested participants, we plan to elaborate on the workshop's discussions in a comprehensive journal article. A prospective publication venue for this extended discourse is "ACM Transactions on Computer-Human Interaction", aiming to contribute significantly to the field's body of knowledge.

5 RECRUITMENT

We are initiating a recruitment process to cultivate a diverse and dynamic group of participants for our half-day workshop on the control-corporeality framework. Our goal is to bring together up to 36 individuals passionate about exploring the nuances of shared control in human-computer interaction and the implications of body-actuating technologies across various domains. The workshop will be advertised across several research communities, including CHI, UIST, ASSETS, Augmented Humans, and MobileHCI, leveraging our extensive research network. Promotion will also extend to the workshop website and social media platforms to ensure wide visibility. Interested applicants should submit their responses to this [Google form](#) by 25th March 2024. Acceptance notifications will be sent by 30th March 2024, along with further details about the workshop schedule and preparation.

6 ORGANISERS

- **Rakesh Patibanda**, a final year PhD student at Monash University's Exertion Games Lab, Melbourne, Australia specialises in the intersection of human-computer interaction

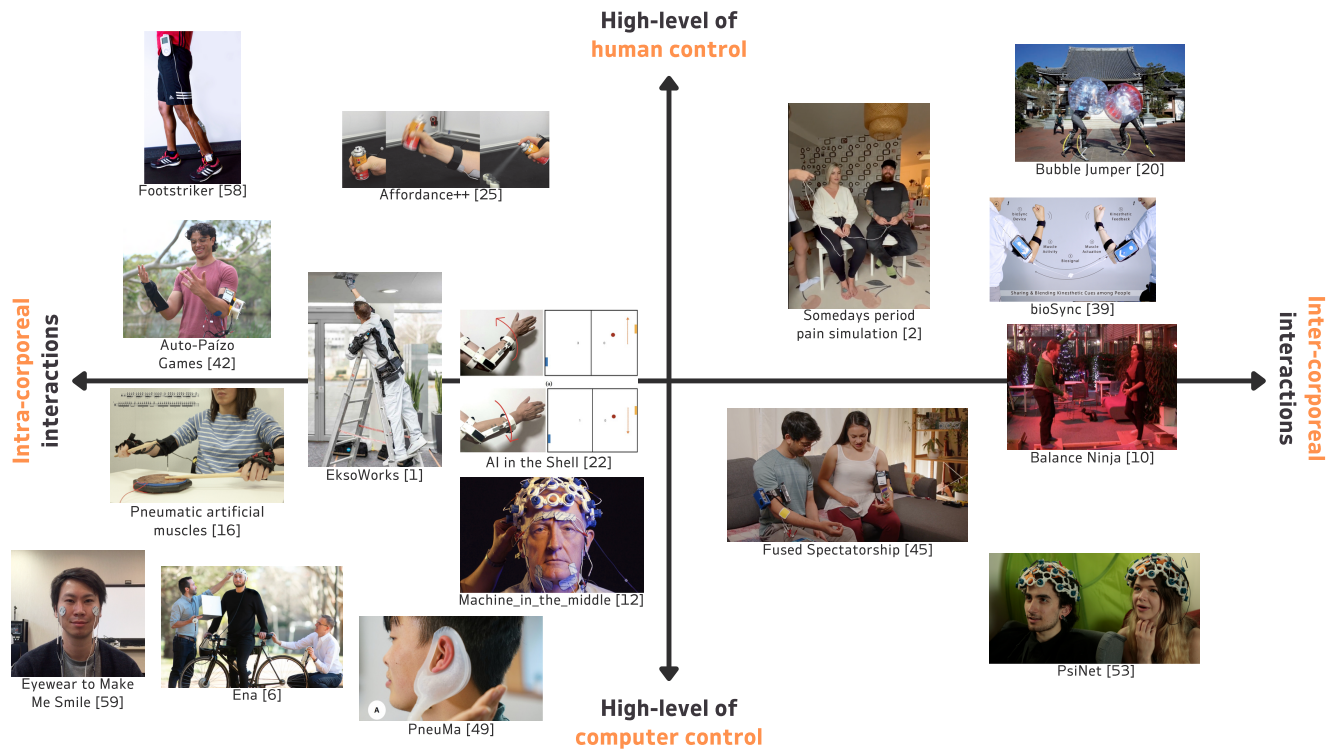


Figure 1: The Control-Corporeality Framework, illustrating examples of human augmentation systems arrayed across a spectrum from high-level human control to high-level computer control and from intra-corporeal to inter-corporeal interactions.

and bodily games. His research explores the innovative domain of body-actuating technologies, particularly Electrical Muscle Stimulation (EMS), and its applications in transforming the human body into an interactive output display. This work is crucial for understanding how shared bodily control can enhance the experience of spectating and playing games [43, 44, 46, 47].

- **Nathalie Overdeest**, a second-year PhD student at the Exertion Games Lab, Monash University, Melbourne, Australia. With a keen interest in human-technology relations, she investigates how the field of HCI can design technologies that support remembering through bodily movement. Her work includes cross-modal interactions, where gaze aversion and vocalisations are used to actuate on-body interfaces, like actuators worn on the skin and exo-skeletons. Her prototypes play around with sensory perception and memory, emphasising proprioception and interoception as part of the remembering experience [3, 40, 41].
- **Aryan Saini** is a 2nd-year PhD student at the Exertion Games Lab. His research focuses on building and evaluating user experiences around bodily extensions that promote embodiment in everyday life [48–50]. Aryan’s research interests include soft exoskeletons, embodied interactions, and wearables. He has previous experience as a research assistant at Microsoft Research India and Weave Lab, IIIT-Delhi.
- **Zhuying Li** is an Associate Professor at the School of Computer Science and Engineering, Southeast University (SEU), China. She is interested in developing humanised technology

that helps people to be more attuned to themselves, others, and the world. In particular, she has researched Human-computer Integration (HInt), investigating how bodily-integrated technologies such as ingestibles can be designed to be interwoven with everyday life to augment one’s daily experience [21, 22, 27, 31].

- **Josh Andres** is a human-computer interaction (HCI), interaction design, user experience researcher and educator. He is currently a Senior Lecturer in the School of Cybernetics at the Australian National University (ANU), investigating the design, experiential dimensions, and futures of emerging technologies, especially interactions with and enabled by intelligent computation such as AI, ubiquitous environments, human-machine co-pilot experiences, and experiences to support wellbeing [4–7, 13].
- **Jarrod Knibbe** is a researcher in Interaction Design. He enjoys building new interactive devices and understanding the experience of using those devices [9, 15, 19]. Currently, he has a special interest in Haptics for Extended Reality and Smart Clothing for Muscle Control. He would describe himself as a future technologist and pursues technology-first human-computer interaction, revealing and exploring novel technologies that may later provide solutions to human-centered problems.
- **Elise van den Hoven** is a Professor of Human-Computer Interaction at the University of Technology Sydney and Eindhoven University of Technology, and is an Honorary

Professor at the University of Dundee. Elise’s research interests span different disciplines, including interaction design, human-computer interaction, and cognitive psychology, which all come together in the research program she leads called Materialising Memories [55–57].

- **Florian ‘Floyd’ Mueller** is the director of the Exertion Games Lab and Professor of Future Interfaces at Monash University, Melbourne, Australia. At the Exertion Games Lab, he investigates the future of bodily play, researching the intersection of technology, the human body and play more broadly [30, 32, 34, 35].

ACKNOWLEDGMENTS

Rakesh Patibanda, Aryan Saini, Nathalie Overdevest, Elise van den Hoven and Florian ‘Floyd’ Mueller thank the Australian Research Council (Discovery Project Grant - DP190102068). Florian ‘Floyd’ Mueller also thanks the Australian Research Council for DP200102612 and LP210200656.

REFERENCES

- [1] [n. d.]. <https://eksobionics.com/eksoworks/>. Accessed: 2024-02-21.
- [2] [n. d.]. @thebirdspapaya “feels like early stage labour”. TikTok. <https://www.tiktok.com/@getsomedays/video/7127827376255307050>. Accessed: 2024-02-21.
- [3] Maysoon Al-quaran, Dominique Geißler, Ceri Kamoan, Nathalie Overdevest, Marijke van Lune, Iris Weijers, and Ruben Gouveia. 2022. MoodTurner: A Self-Tracking Smart Jewellery to Support Awareness and Reflection in Sensory Processing Sensitivity Self-Care. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA ’22). Association for Computing Machinery, New York, NY, USA, Article 432, 6 pages. <https://doi.org/10.1145/3491101.3519803>
- [4] Josh Andres, Julian de Hoog, and Florian ‘Floyd’ Mueller. 2018. “I Had Super-Powers When EBike Riding” Towards Understanding the Design of Integrated Exertion. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play* (Melbourne, VIC, Australia) (CHI PLAY ’18). Association for Computing Machinery, New York, NY, USA, 19–31. <https://doi.org/10.1145/3242671.3242688>
- [5] Josh Andres, m.c. schraefel, Rakesh Patibanda, and Florian ‘Floyd’ Mueller. 2020. Future InBodied: A Framework for Inbodied Interaction Design. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction* (TEI ’20). Association for Computing Machinery, New York, NY, USA, 885–888. <https://doi.org/10.1145/3374920.3374969>
- [6] Josh Andres, m.c. schraefel, Nathan Semertzidis, Brahmi Dwivedi, Yutika C. Kulwe, Juerg von Kaenel, and Florian Floyd Mueller. 2020. Introducing Peripheral Awareness as a Neurological State for Human-computer Integration. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, Hawaii, USA) (CHI ’20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376128>
- [7] Josh Andres, Alexandra Zafiroglu, Katherine Daniell, Paul Wong, Mina Henein, Xuanying Zhu, Ben Sweeting, Michael Arnold, Delia Pembrey Macnamara, and Ariella Helfgott. 2023. Cybernetic Lenses for Designing and Living in a Complex World. In *Proceedings of the 34th Australian Conference on Human-Computer Interaction* (Canberra, ACT, Australia) (OzCHI ’22). Association for Computing Machinery, New York, NY, USA, 348–351. <https://doi.org/10.1145/3572921.3576209>
- [8] Steve Benford, Richard Ramchurn, Joe Marshall, Max L. Wilson, Matthew Pike, Sarah Martindale, Adrian Hazzard, Chris Greenhalgh, Maria Kallionpää, Paul Tennent, and Brendan Walker. 2021. Contesting control: journeys through surrender, self-awareness and looseness of control in embodied interaction. *Human-Computer Interaction* 36, 5–6 (Oct 2021), 361–389. <https://doi.org/10.1080/07370024.2020.1754214>
- [9] Joanna Bergström, Jarrod Knibbe, Henning Pohl, and Kasper Hornbæk. 2022. Sense of Agency and User Experience: Is There a Link? *ACM Transactions on Computer-Human Interaction* 29, 4 (March 2022), 28:1–28:22. <https://doi.org/10.1145/3490493>
- [10] Richard Byrne, Joe Marshall, and Florian ‘Floyd’ Mueller. 2016. Balance Ninja: Towards the Design of Digital Vertigo Games via Galvanic Vestibular Stimulation. In *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play* (Austin, Texas, USA) (CHI PLAY ’16). Association for Computing Machinery, New York, NY, USA, 159–170. <https://doi.org/10.1145/2967934.2968080>
- [11] Roger Cailliois. 2001. *Man, Play, and Games*. University of Illinois Press. Google-Books-ID: bDjOPsjzfc4C.
- [12] Rod Dickinson, Nathan Semertzidis, and Florian Floyd Mueller. 2022. Machine In The Middle: Exploring Dark Patterns of Emotional Human-Computer Integration Through Media Art. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA ’22). Association for Computing Machinery, New York, NY, USA, Article 43, 7 pages. <https://doi.org/10.1145/3491101.3503555>
- [13] Juerg von Kaenel Florian ‘Floyd’ Mueller, Jonathan Duckworth and Josh Andres. 2023. Taking inspiration from becoming “one with a bike” to design human-computer integration. *Human-Computer Interaction* 0, 0 (2023), 1–26. <https://doi.org/10.1080/07370024.2023.2269148> Citation Key: doi:10.1080/07370024.2023.2269148 tex.eprint: <https://doi.org/10.1080/07370024.2023.2269148>
- [14] Thomas Fuchs. 2017. *Intercorporeality and Interactivity*. Oxford University Press, 0. <https://doi.org/10.1093/acprof:oso/9780190210465.003.0001>
- [15] Graeme Gange and Jarrod Knibbe. 2021. Optimising Automatic Calibration of Electric Muscle Stimulation. *Proceedings of the AAAI Conference on Artificial Intelligence* 35, 55 (May 2021), 3759–3767. <https://ojs.aaai.org/index.php/AAAI/article/view/16493>
- [16] Takashi Goto, Swagata Das, Katrin Wolf, Pedro Lopes, Yuichi Kurita, and Kai Kunze. 2020. Accelerating Skill Acquisition of Two-Handed Drumming using Pneumatic Artificial Muscles. In *Proceedings of the Augmented Humans International Conference* (Kaiserslautern, Germany) (AHs ’20). Association for Computing Machinery, New York, NY, USA, Article 12, 9 pages. <https://doi.org/10.1145/3384657.3384780>
- [17] Johan Huizinga. 2016. *Homo Ludens: A Study of the Play-Element in Culture*. Angelico Press, Kettering, OH.
- [18] Katherine Isbister and Florian ‘Floyd’ Mueller. 2015. Guidelines for the Design of Movement-Based Games and Their Relevance to HCI. *Human-Computer Interaction* 30, 3–4 (May 2015), 366–399. <https://doi.org/10.1080/07370024.2014.996647>
- [19] Jarrod Knibbe, Adrian Alsmith, and Kasper Hornbæk. 2018. Experiencing Electrical Muscle Stimulation. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 3 (Sep 2018). <https://doi.org/10.1145/3264928>
- [20] Kai Kunze, Kouta Minamizawa, Stephan Lukosch, Masahiko Inami, and Jun Rekimoto. 2017. Superhuman Sports: Applying Human Augmentation to Physical Exercise. *IEEE Pervasive Computing* 16, 2 (2017), 14–17. <https://doi.org/10.1109/MPRV.2017.35>
- [21] Xiang Li, Xiaohang Tang, Xin Tong, Rakesh Patibanda, Florian ‘Floyd’ Mueller, and Hai-Ning Liang. 2021. Myopic Bike and Say Hi: Games for Empathizing with The Myopic. In *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play* (CHI PLAY ’21). Association for Computing Machinery, New York, NY, USA, 333–338. <https://doi.org/10.1145/3450337.3483505>
- [22] Zhuying Li, Tianze Huang, Rakesh Patibanda, and Florian Mueller. 2023. AI in the Shell: Towards an Understanding of Integrated Embodiment. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA ’23). Association for Computing Machinery, New York, NY, USA, Article 28, 7 pages. <https://doi.org/10.1145/3544549.3585867>
- [23] Pedro Lopes and Patrick Baudisch. 2017. Immense Power in a Tiny Package: Wearables Based on Electrical Muscle Stimulation. *IEEE Pervasive Computing* 16, 3 (2017), 12–16. <https://doi.org/10.1109/MPRV.2017.2940953>
- [24] Pedro Lopes, Alexandra Ion, Willi Mueller, Daniel Hoffmann, Patrik Jonell, and Patrick Baudisch. 2015. Proprioceptive Interaction. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (CHI ’15). Association for Computing Machinery, New York, NY, USA, 939–948. <https://doi.org/10.1145/gj37np>
- [25] Pedro Lopes, Patrik Jonell, and Patrick Baudisch. 2015. Affordance++: Allowing Objects to Communicate Dynamic Use. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (CHI ’15). Association for Computing Machinery, New York, NY, USA, 2515–2524. <https://doi.org/10.1145/2702123.2702128>
- [26] Yash Dhanpal Mehta, Rohit Ashok Khot, Rakesh Patibanda, and Florian ‘Floyd’ Mueller. 2018. Arm-A-Dine: Towards Understanding the Design of Playful Embodied Eating Experiences. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play* (CHI PLAY ’18). Association for Computing Machinery, New York, NY, USA, 299–313. <https://doi.org/10.1145/3242671.3242710>
- [27] Maria F. Montoya, Rakesh Patibanda, Christal Clashing, Sarah Jane Pell, and Florian ‘Floyd’ Mueller. 2022. Towards an Initial Understanding of the Design of Playful Water Experiences Through Flotation. In *Extended Abstracts of the 2022 Annual Symposium on Computer-Human Interaction in Play* (CHI PLAY ’22). Association for Computing Machinery, New York, NY, USA, 120–126. <https://doi.org/10.1145/3505270.3558324>
- [28] James W. Moore. 2016. What Is the Sense of Agency and Why Does it Matter? *Frontiers in Psychology* 7 (2016). <https://www.frontiersin.org/article/10.3389/fpsyg.2016.01272>
- [29] Florian Mueller and Katherine Isbister. 2014. Movement-based game guidelines. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI ’14). Association for Computing Machinery, New York, NY, USA, 2191–2200. <https://doi.org/10.1145/ggcj93>

- [30] Florian 'Floyd' Mueller, Richard Byrne, Josh Andres, and Rakesh Patibanda. 2018. Experiencing the Body as Play. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173784>
- [31] Florian 'Floyd' Mueller, Tuomas Kari, Zhuying Li, Yan Wang, Yash Dhanpal Mehta, Josh Andres, Jonathan Marquez, and Rakesh Patibanda. 2020. Towards Designing Bodily Integrated Play. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '20)*. Association for Computing Machinery, New York, NY, USA, 207–218. <https://doi.org/10.1145/3374920.3374931>
- [32] Florian 'Floyd' Mueller, Rakesh Patibanda, Richard Byrne, Zhuying Li, Yan Wang, Josh Andres, Xiang Li, Jonathan Marquez, Stefan Greuter, Jonathan Duckworth, and Joe Marshall. 2021. Limited Control Over the Body as Intriguing Play Design Resource. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3411764.3445744>
- [33] Florian 'Floyd' Mueller, Nathan Semertzidis, Josh Andres, Joe Marshall, Steve Benford, Xiang Li, Louise Matjeka, and Yash Mehta. 2023. Towards Understanding the Design of Intertwined Human-Computer Integrations. *ACM Trans. Comput.-Hum. Interact.* (apr 2023). <https://doi.org/10.1145/3590766> Just Accepted.
- [34] Florian 'Floyd' Mueller, Nathan Semertzidis, Josh Andres, Martin Weigel, Suranga Nanayakkara, Rakesh Patibanda, Zhuying Li, Paul Strohmeier, Jarrod Knibbe, Stefan Greuter, Marianna Obrist, Pattie Maes, Dakuo Wang, Katrin Wolf, Liz Gerber, Joe Marshall, Kai Kunze, Jonathan Grudin, Harald Reiterer, and Richard Byrne. 2022. Human-Computer Integration: Towards Integrating the Human Body with the Computational Machine. *Foundations and Trends® in Human-Computer Interaction* 16, 1 (Oct 2022), 1–64. <https://doi.org/10.1561/11000000086>
- [35] Florian 'Floyd' Mueller, Martin R. Gibbs, Frank Vetere, and Darren Edge. 2017. Designing for Bodily Interplay in Social Exertion Games. *ACM Transactions on Computer-Human Interaction* 24, 3 (May 2017), 24:1–24:41. <https://doi.org/10/gj6p7q>
- [36] Junichi Nabeshima, M. H. D. Yamen Saraiji, and Kouta Minamizawa. 2019. Prosthetic tail artificial anthropomorphic tail for extending innate body functions. In *Proceedings of the 10th Augmented Human International Conference, AH 2019*. Association for Computing Machinery, a36. <https://doi.org/10.1145/3311823.3311848>
- [37] Suranga Chandima Nanayakkara, Masahiko Inami, Florian Mueller, Jochen Huber, Chitralekha Gupta, Christophe Jouffrais, Kai Kunze, Rakesh Patibanda, Samantha W T Chan, and Moritz Alexander Messerschmidt. 2023. Exploring the Design Space of Assistive Augmentation. In *Proceedings of the Augmented Humans International Conference 2023 (Glasgow, United Kingdom) (AHs '23)*. Association for Computing Machinery, New York, NY, USA, 371–373. <https://doi.org/10.1145/3582700.3582729>
- [38] Nguyen. 2020. *Games: Agency As Art*. Oxford University Press USA, New York, NY.
- [39] Jun Nishida and Kenji Suzuki. 2017. bioSync: A Paired Wearable Device for Blending Kinesthetic Experience. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (Denver, Colorado, USA) (CHI '17)*. Association for Computing Machinery, New York, NY, USA, 3316–3327. <https://doi.org/10.1145/3025453.3025829>
- [40] Nathalie Overvest, Rakesh Patibanda, Aryan Saini, Elise van den Hoven, and Florian 'Floyd' Mueller. 2024. GazeAway: Designing for Gaze Aversion Experiences. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (Honolulu, Hawaii) (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3613905.3650771>
- [41] Nathalie Overvest, Rakesh Patibanda, Aryan Saini, Elise Van Den Hoven, and Florian 'Floyd' Mueller. 2023. Towards Designing for Everyday Embodied Remembering: Findings from a Diary Study. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (Pittsburgh, PA, USA) (DIS '23)*. Association for Computing Machinery, New York, NY, USA, 2611–2624. <https://doi.org/10.1145/3563657.3595999>
- [42] Rakesh Patibanda, Chris Hill, Aryan Saini, Xiang Li, Yuzheng Chen, Andrii Matvienko, Jarrod Knibbe, Elise van den Hoven, and Florian 'Floyd' Mueller. 2023. Auto-Paizo Games: Towards Understanding the Design of Games That Aim to Unify a Player's Physical Body and the Virtual World. *Proc. ACM Hum.-Comput. Interact.* 7, CHI PLAY, Article 408 (oct 2023), 26 pages. <https://doi.org/10.1145/3611054>
- [43] Rakesh Patibanda, Xiang Li, Yuzheng Chen, Aryan Saini, Christian N Hill, Elise van den Hoven, and Florian 'Floyd' Mueller. 2021. Actuating Myself: Designing Hand-Games Incorporating Electrical Muscle Stimulation. In *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '21)*. Association for Computing Machinery, New York, NY, USA, 228–235. <https://doi.org/10.1145/3450337.3483464>
- [44] Rakesh Patibanda, Florian 'Floyd' Mueller, Matevz Leskovsek, and Jonathan Duckworth. 2017. Life Tree: Understanding the Design of Breathing Exercise Games. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '17)*. Association for Computing Machinery, New York, NY, USA, 19–31. <https://doi.org/10.1145/3116595.3116621>
- [45] Rakesh Patibanda, Aryan Saini, Nathalie Overvest, Maria F. Montoya, Xiang Li, Yuzheng Chen, Shreyas Nisal, Josh Andres, Jarrod Knibbe, Elise van den Hoven, and Florian 'Floyd' Mueller. 2023. Fused Spectatorship: Designing Bodily Experiences Where Spectators Become Players. *Proceedings of the ACM on Human-Computer Interaction* 7, CHI PLAY (Oct. 2023), 403:769–403:802. <https://doi.org/10.1145/3611049>
- [46] Rakesh Patibanda, Nathan Arthur Semertzidis, Michaela Scary, Joseph Nathan La Delfa, Josh Andres, Mehmet Aydin Baytaş, Anna Lisa Martin-Niedecken, Paul Strohmeier, Bruno Fruchard, Sang-won Leigh, et al. 2020. Motor Memory in HCI. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3375163>
- [47] Rakesh Patibanda, Elise Van Den Hoven, and Florian 'Floyd' Mueller. 2022. Towards Understanding the Design of Body-Actuated Play. In *Extended Abstracts of the 2022 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '22)*. Association for Computing Machinery, New York, NY, USA, 388–391. <https://doi.org/10.1145/3505270.3558367>
- [48] Aryan Saini, Haotian Huang, Rakesh Patibanda, Nathalie Overvest, Elise Van Den Hoven, and Florian 'Floyd' Mueller. 2022. SomaFlatables: Supporting Embodied Cognition through Pneumatic Bladders. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3526114.3558705>
- [49] Aryan Saini, Rakesh Patibanda, Nathalie Overvest, Elise Van Den Hoven, and Florian 'Floyd' Mueller. 2024. PneuMa: Designing Pneumatic Bodily Extensions for Supporting Movement in Everyday Life. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction (Cork, Ireland) (TEI '24)*. Association for Computing Machinery, New York, NY, USA, Article 1, 16 pages. <https://doi.org/10.1145/3623509.3633349>
- [50] Aryan Saini, Srihari Sridhar, Aarushi Raheja, Rakesh Patibanda, Nathalie Overvest, Po-Yao (Cosmos) Wang, Elise Van Den Hoven, and Florian Floyd Mueller. 2023. Pneucocchio: A playful nose augmentation for facilitating embodied representation. In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (San Francisco, CA, USA) (UIST '23 Adjunct)*. Association for Computing Machinery, New York, NY, USA, Article 40, 3 pages. <https://doi.org/10.1145/3586182.3616651>
- [51] Katie Salen and Eric Zimmerman. 2004. *Rules of Play: Game Design Fundamentals*. MIT Press.
- [52] M. H. D. Yamen Saraiji, Tomoya Sasaki, Kai Kunze, Kouta Minamizawa, and Masahiko Inami. 2018. MetaArmS: Body remapping using feet-controlled artificial arms. In *UIST 2018 - Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, Inc, 65–74. <https://doi.org/10.1145/3242587.3242665>
- [53] Nathan Semertzidis, Michaela Jayne Vranic-Peters, Xiao Zoe Fang, Rakesh Patibanda, Aryan Saini, Don Samitha Elvitigala, and Florian 'Floyd' Mueller. 2024. PsiNet: Toward Understanding the Design of Brain-to-Brain Interfaces for Augmenting Inter-Brain Synchrony. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (Honolulu, Hawaii) (CHI '24)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3613904.3641983>
- [54] Nathan Arthur Semertzidis, Michaela Vranic-Peters, Xiao Fang, Xinyi Wang, Rakesh Patibanda, Josh Andres, Paul Strohmeier, Kai Kunze, Pedro Lopes, Fabio Zambetta, and Florian Floyd Mueller. 2021. SIGHInt: Special Interest Group for Human-Computer Integration. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (Yokohama, Japan) (CHI EA '21)*. Association for Computing Machinery, New York, NY, USA, Article 163, 3 pages. <https://doi.org/10.1145/3411763.3450400>
- [55] Elise van den Hoven. 2014. A future-proof past: Designing for remembering experiences. *Memory Studies* 7, 3 (July 2014), 370–384. <https://doi.org/10.1177/1750698014530625>
- [56] Elise van den Hoven and Berry Eggen. 2008. Informing augmented memory system design through autobiographical memory theory. *Personal and Ubiquitous Computing* 12, 6 (Aug. 2008), 433–443. <https://doi.org/10.1007/s00779-007-0177-9>
- [57] Elise van den Hoven, Joep Frens, Dima Aliakseyeu, Jean-Bernard Martens, Kees Overbeeke, and Peter Peters. 2007. Design research & tangible interaction. In *Proceedings of the 1st international conference on Tangible and embedded interaction (TEI '07)*. Association for Computing Machinery, New York, NY, USA, 109–115. <https://doi.org/10.1145/1226969.1226993>
- [58] Frederik Wiehr, Felix Kosmalla, Florian Daiber, and Antonio Krüger. 2017. Foot-Striker: An EMS-Based Foot Strike Assistant for Running. In *Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers (UbiComp '17)*. Association for Computing Machinery, New York, NY, USA, 317–320. <https://doi.org/10.1145/3123024.3123191> event-place: Maui, Hawaii.
- [59] Lai Yen-Chin, YuanLing Feng, Kai Kunze, Junich Shimizu, and Takuro Nakao. 2017. Eyewear to Make Me Smile: Can Electric Muscle Stimulation Increase Happiness?. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction (Yokohama, Japan) (TEI '17)*. Association for Computing Machinery, New York, NY, USA, 579–582. <https://doi.org/10.1145/>

3024969.3025097